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ScienceDirect

Procedia Computer Science 259 (2025) 140–149

Procedia Computer Science

www.elsevier.com/locate/procedia

Sixth International Conference on Futuristic Trends in Networks and Computing Technologies (FTNCT06) held in Uttarakhand, India

Advanced AI-Powered Intrusion Detection Systems in Cybersecurity Protocols for Network Protection

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Abstract

Conventional rule-based network intrusion detection systems (NIDS) find it difficult to remain with the increasing complexity of cyber-attacks. To solve these issues, this study examines the development of NIDS as well as the transformative potential of artificial intelligence (AI). AI-enhanced NIDS can efficiently identify and respond to known and unknown threats in real-time by utilizing machine learning (ML) techniques. The system can differentiate between typical network behavior and abnormalities using both supervised and unsupervised learning techniques, as opposed to depending exclusively on pre-established rules. The accuracy and adaptability of the system are further improved by deep learning (DL) architectures like recurrent neural networks (RNNs) and convolutional neural networks (CNNs). The paper explores the past developments of intrusion detection, comparing rule-based approaches to modern AI-driven systems. It discusses cutting-edge techniques like anomaly detection, ensemble methods, and hybrid models. While Recognizing issues such as adversarial attacks and interpretability, the article underlines the importance of AI-enhanced NIDS in protecting digital infrastructure. This study provides a complete overview, unique insights, and practical advice for cybersecurity experts looking to install and optimize AI-powered intrusion detection solutions.

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Peer-review under responsibility of the scientific committee of the Sixth International Conference on Futuristic Trends in Networks and Computing Technologies (FTNCT06)

Keywords: Cybersecurity; Artificial Intelligence; Intrusion Detection Systems; Network Security; NIDS.

1. Introduction

The possibilities for cyber threats are inexhaustible given that organizations rely on interconnected networks for smooth operations. Therefore, the defense system must be not only strong but also smart to counter the continuous changes in the nature of the cyberattacks and constantly expanding attack surface. It's where the AI-powered Network Intrusion Detection System (NIDS) comes in, the latest weapon in the battle against cyber criminals [1]. The traditional security solutions cease to be effective measures of protection against cyber attackers due to the ever-developing techniques used by the criminals. The necessity of enhanced approaches that would enable

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Peer-review under responsibility of the scientific committee of the Sixth International Conference on Futuristic Trends in Networks and Computing Technologies (FTNCT06)
10.1016/j.procs.2025.03.315

preventing cyber-attacks from happening in the first place has never been as critical as it is now [2]. This article is aimed at presenting the reader with information about cybersecurity as a promising and constantly developing industry, as well as the potential of AI-powered NIDS in enhancing the protection of today's organizations. The problems associated with Traditional IDS include its inability to provide timely responses to incidents and it has a high number of False Positives. However, in IDS systems that are powered by Artificial Intelligence, these challenges are averted since the system works according to vast data collected in order to identify the normal network traffic patterns. They are capable of detecting subverted deviations from this norm even to the extent of discerning new and unencountered forms of attacks. AI skills for signify learning and pattern recognition are effective to improve threat identification, isolation and subsequent measures that take the burden off the cybersecurity professionals and increase overall cybersecurity [3]. The type of danger that organizations are experiencing in the sphere of cyberspace cannot be stopped with the classical security systems. APT, ransomware and zero-day attacks have raised the questions about the necessity of a shift to a more proactive approach to cybersecurity. The problem with current IDS systems and their counterparts, legacy systems, is that they use a series of rules described in signatures to detect and analyze the network traffic. As such, these older systems are not so proficient in identifying variations from vectors of known form and, therefore, are more likely to produce false positives while at the same time missing many threats. In an environment where delayed responses to potential intrusions can be catastrophic, the importance of real-time threat detection and immediate, targeted responses cannot be overstated [4]. AI-powered NIDS leverage cutting-edge machine learning algorithms and deep learning models to identify complex patterns and anomalies as they occur, providing a resilient and dynamic defense against the constantly shifting strategies of cyber adversaries.

2. Related Work

Delving into the current state of research and existing solutions is paramount for developing an effective AI-powered Network Intrusion Detection System (NIDS). Researchers and cybersecurity professionals have been actively exploring diverse methodologies to enhance the efficacy of intrusion detection systems, leveraging the latest breakthroughs in artificial intelligence and machine learning [5]. A comprehensive examination of related work sheds light on the evolution of NIDS and provides invaluable insights into the cutting edge of this field. NIDS most of its origins from traditional Intrusion Detection Systems which involves the use of rules and signatures to detect intrusions. They are still somewhat efficient at defending against known threats, but thoroughly ineffective against new forms of attacks. The study in this area was first directed toward the development of new and optimized databases containing signatures and toward refining the rule-based engines in order to achieve higher levels of accuracy in detecting intrusions [6]. One of the advancements that was made in NIDS was the integration of machine learning into the design of what was already known as NIDS. First approaches were oriented on including known algorithms of supervised learning, based on the decision tree and support vector machines, combining them with an attempt of training of models on the initial labeled data sets [7]. These models could then be able to distinguish normal and anomalous behavior for the local area network and hence offer a more kinetic solution. In the further process of development, since the threat became more sophisticated, the center of attention was the statistical anomaly detection adopted by the unsupervised learning algorithms. This approach brought in the ability for systems to monitor for any abnormality without actually having to rely on specific attack signatures, and thus new threats could be easily discovered [8]. It must also be noted that the story of NIDS has changed greatly since the arrival of deep learning. CNNs as well as RNNs have proved to be very capable of identifying intricate patterns in big data. NIDS has benefitted from these models' capability, thus enabling the conducting of feature extraction from raw data with higher efficiency and accuracy. Hybrid and Ensemble methods are used when two or more algorithms are combined it in a systematic way so that it produces a better performance compared to when they are used individually. New studies have appeared recently that focus on the development of a multiple layered approach that uses both signature-based, anomaly-based, and machine learning methods to construct stronger NIDS [9]. The use of ensemble methods where two or more models are combined with the intention of boosting the general detection capability is perhaps a way of portraying the increasing need for holistic approaches in mitigating various kinds of threats in the cyber domain. However, some issues still arise when implementing AI-based NIDS, there are still some hitches when implementing AI based NIDS.

3. Materials and methods

Conventionally used Intrusion Detection Systems (IDS) have several limitations that make it difficult for them to counter modern threats. However, one major point of criticism is that they are not very capable of responding to incidents rapidly. Even with a threat detected, an organization may fumble on the best line of action hence the potential of security breaches. Moreover, traditional IDS generates high rate of false alarms which creates a problem due to wasting resources and time, while real security threats are hidden deep down. Performing conventional network traffic analysis has limitations due to the vast data collected, and this has been solved by using Machine learning and Deep learning in Network Intrusion Detection Systems (NIDS) which analyses big data to identify unusual patterns that could be a signal of a cyberattack [10]. This is because it is a more enhanced form of threat detection that helps organizations eliminate threats as and when they appear in a bid to cause so much havoc to the entire system.

The major strength surrounding AI-based cyber threat detection is the ability to analyze and recognize normal behaviour within an organization's network and cloud services. While previous solutions encode static behaviors of users, analyzing them against predetermined patterns, AI records and constantly assesses disturbances of specific user's behavior and receives validation from security analysts [11]. This approach solves such threats as privilege escalation, ransomware, data exfiltration, lateral movement, command-and-control server connectivity, and zero-day attacks. Figure 1(a) illustrates the Artificial intelligence based intrusion detection system.

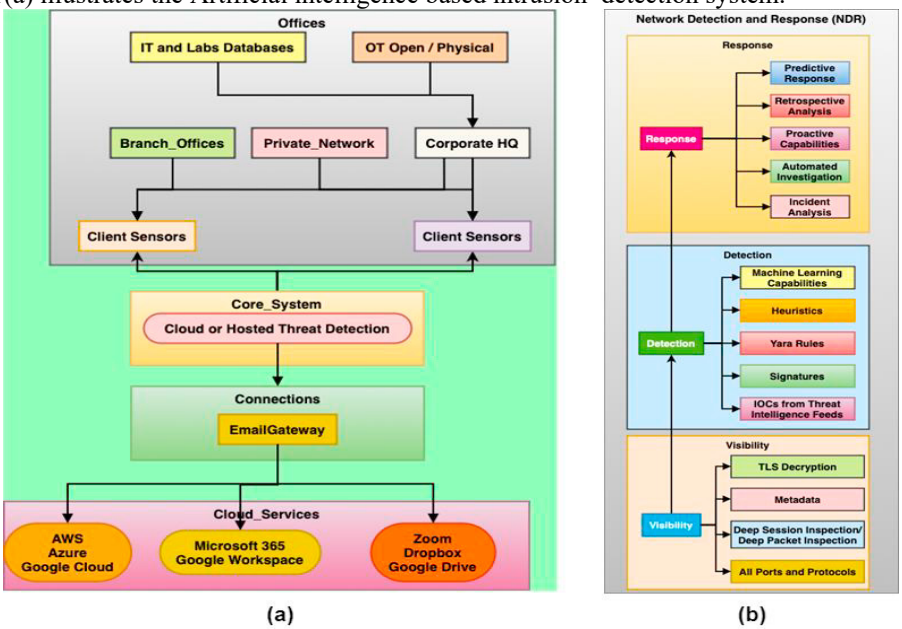

Figure 1. (a) AI-based Intrusion detection system (b) Schema of basic NDR

AI integrated NIDS is a new evolutionary step that brings hope in cyber security warfare against cyber criminals. These systems create a better proactive and responsive system for cybersecurity which is essential to the protection of data and a way of preventing losses. Overall, AI-powered NIDS are a powerful tool for organizations that want to improve their cybersecurity posture. They offer a number of benefits over traditional IDS, and they are a critical part of any modern cybersecurity strategy. In the current era of advanced technology and globalization, organizations are encountering multifaceted and complex cyber threats that are beyond the ability of traditional security mechanisms [12]. From independent cyber criminals to organized state directed personnel, cyber attackers are improving on their approaches, putting more pressure on traditional security architectures. APT, ransomware attacks, and zero-day features demand a creative approach that will be the foundation of an effective cybersecurity program. The primary elements making it possible to identify a cyber threat in real time and respond to it promptly are essential in protecting organizations. It makes it possible for security teams to respond promptly and address

threats as they surface so as to minimize the impact on an organization and would mean less likelihood of loss of data [13]. AI offers a progressive architecture of defense which is capable of learning and improving on its functions and capable of detecting new and complicated patterns and behavior in real time [9]. The growth of networks beyond cloud systems and the increasing intricateness of structures have provided opportunities for cybercriminals. The nature of distributed networks where big amount of data is processed can be easily camouflaged with malicious activities [14]. Concerning this, NDR systems solve it by collecting data from different network devices and using analytics like machine learning to detect threats that other security tools may overlook.

The Network Detection and Response (NDR) systems provide a round-the-clock surveillance of the entire network structure with diverse elements including data centers, cloud platforms, On-premises devices, and remote workstations. The schema of the network detection response (NDR) system and its basic building blocks are shown in Figure 1(b). This end-to-end visibility is critical in today's dispersed environment where users, devices and services are located in different places. Behavioral analytics and AI are other related concepts, where the first refers to the analysis of user behavior patterns, while the second – is the use of artificial intelligence technologies. The AI-driven NDR systems expand the capability of a SOC since it automatically discovers security events. These systems not only identify hard-to-detect and concealing attacks in the networks, but also have the capability to counter such attacks in real-time as and when they occur. Further, it can be said that while using NDR systems, organizations might have limited visibility of the cloud data or other identity security-related data [15]. Some of the NDR tools can also be isolated in terms of consoles, making the security analysts to work across different console thereby causing interruption of threat investigation and response.

3.1. The architecture of AI-Powered NIDS

The element of artificial intelligence in NIDS is also embedded into a multi-tiered model that augments the traditional NIDS structure[16].Network data is collected from web traffic log files, endpoints, as well as sensors. This raw data is further processed so as to select the important features for creation of the models of a machine learning system. Most of the machine learning models require data in a format and structure which makes it easier for them to be processed, and this is where the key features are extracted from the preprocessed data by the AI algorithms. The focus of an AI-based NIDS is the operational use of machine and deep learning algorithms to analyze the vast data gathering source from which the data is trained and parsed down into normal network traffic and suspicious patterns. Network traffic is also frequently analyzed with machine learning models; the primary goal of which is not to detect threats but rather check whether specific patterns similar to threats have been observed in the network traffic. AI in NIDS are developed in such a way that they adapt to changes in the environment. It means that they are constantly evolving from the data obtained from more modern and more complex cyber threats.

3.2. Machine learning algorithms in AI-powered NIDS

A diverse range of machine learning algorithms are employed in AI-powered NIDS, each offering unique strengths and capabilities [17]. Some of the methods, such as Support Vector Machines (SVMs), decision trees, and random forests are learned from the labeled data and help the system to properly classify the network activity. Some of the techniques that are used in detecting anomalous behaviour in a network include k-means clustering and Gaussian Mixture Models (GMMs) where no prior knowledge of the attack signatures is required. This paper differentiates Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) as some of the best in detecting patterns that may exist in network traffic in a manner that enhances the rates of detection.

3.3. Transformative impact of AI-powered NIDS

The integration of AI into NIDS has revolutionized cybersecurity, offering several critical benefits:

- ***Enhanced Threat Detection Accuracy:*** Since APS can detect more threats than signature-based SIEM, it is easier to increase the detection accuracy with NIDS, covering zero-day attacks and unknown methods.
- ***Real-time Threat Detection and Response:*** AI gives the ability to make quick analyses of flow throughout the network and quickly identify threats, hence limiting intrusion damage.

- ***Reduced False Positives:*** AI algorithms are very good in differentiating between normal and malicious behavior hence they give less false alarms that will help security teams in addressing real issues.
- ***Continuous Learning and Adaptation:*** This means that the AI-powered NIDS are constantly modifying it in order to make it suitable for new enemies that may emerge.
- ***Threat Intelligence Integration:*** By integrating the threat intelligence feeds, AI can include context about newly discovered threats and improve the chance of detection.

3.4. Evolution of traditional intrusion detection systems

Conventional IDS has been the cornerstone of network security for a long time and can be mainly categorized into rule-based and signature-based systems. The rule-based systems have been effective in detecting known attack patterns, but have failed for the dynamic or newer and more complex threats [18]. Initial activities in this area were focused on augmenting signature databases, improving rule-based engine, and improving the methods of consolidating and correlating alerts for minimizing false alarms and increasing identification efficacy.

3.5. Machine learning in intrusion detection

IDS improved its security system by incorporating machine learning into it which was a revolution in cybersecurity. Supervised learning methods were employed in training the models on labeled databases so that such systems may discern relations characteristic of both legitimate and tumorous network behaviors. There are several conventional machine learning methods such as decision trees, support vector machines and k-nearest neighbors which were first used to enhance intrusion detection and which offered more dynamism than the traditional systems [19].

3.6. Anomaly detection and unsupervised learning

Cyber threats started growing more diverse and sophisticated and the need to automatically identify these threats occurred due to use of methods based on unsupervised learning. This approach enabled systems to detect the anomaly or the deviant behavior in the network traffic not based on signatures of known attacks. Techniques such as clustering techniques (for example, k-means clustering) and statistical models (for instance, Gaussian Mixture Models) shifted into techniques for identifying new risks and provided the extra layer of protection by predicting changes in the network behavior [20].

3.7. Deep learning approaches

The emergence of deep learning marked a turning point in the approach to intrusion detection. Neural networks especially CNNs and RNNs performed exceptionally well in the identification of several patterns within large database [19]. For the enhanced precision and efficiency of the Network Intrusion Detection Systems (NIDS), deep learning models' capability to extract meaningful features from the raw network data without the need for significant human input was beneficial in increasing the ability of the model to detect the difficult-to-detect threats that the traditional models might fail to identify.

3.8. Ensemble methods and hybrid approaches

In the recent past, the emphasis shifted to developing ensemble techniques, which involves the use of multiple models to up the effectiveness of detection. Such ensemble methods sometimes use several machine learning classifiers and provide a more holistic solution to threat identification. The Block diagram of Ensemble methods and hybrid approaches in NIDS is shown in Figure 2(a). Further, to the main kinds of systems, which combine elements of signature-based systems, anomaly-based systems and finally those based on machine learning, the most effective have been developed, as being a flexible and rather solid antidote to the constantly expanding universe of threats

[17]. This characteristic show how the industry understands that there is need for flexibility and variety in tackling the various problems in cybersecurity that are diverse in nature.

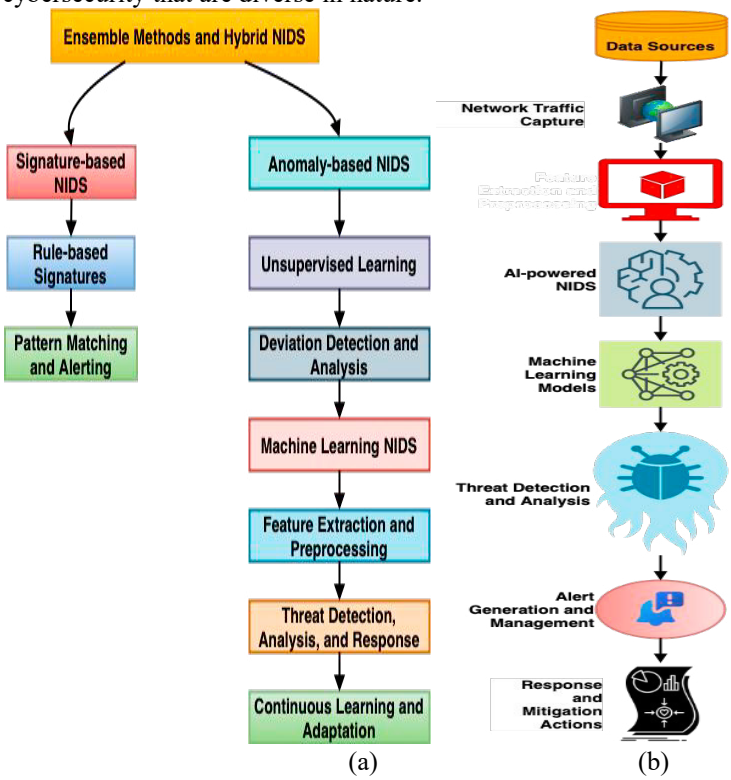

Figure 2. Block diagram of (a) Ensemble methods and hybrid approaches in Network Intrusion Detection Systems (NIDS) (b) Workflow of an AI-Powered NIDS.

As mentioned before ensemble methods use multiple models, traditional or innovative, to improve the detection ability. Since the strengths of each method are accumulated in the models, these methods offer a more complex view on the network traffic analysis, thus making the threat detection more effective and stable [21]. NIDS, Hybrid Network Intrusion Detection Systems are intended to construct a flexible and also robust defense mechanism by incorporating these modes of detection. These systems use the multiple conventional patterns such as, the first one is where the systems look for a specific type of threats with definable fingerprints, the second one is where the systems look for behaviors that are different from normal ones, and lastly the systems use the machine learning approaches that can learn the new emerging patterns [22]. Thus, Signature-based NIDS work by trying to match all the traffic flows with the database of known attacks signatures. Being based on known threats, this method is very efficient in regard to recognizing known threats, though it can miss the new and unusual types of attacks [23]. Anomaly-based NIDS monitor networks through performing data mining and using unsupervised learning algorithms in order to discover abnormalities of traffic patterns. It comes handy when selecting unknown objects, but non-stationary distribution might lead to a high number of false alarms, which in turn should be properly tuned and validated [6]. Strategies of coping with change in an uncertain environment machine learning-based NIDS have been found to use supervised and unsupervised learning in the analysis of the extracted features from network information. These systems have the potential of detecting new and existing threats as long as these systems are periodically updated with new data in order to retrain them for new threats [24]. Since the threat in the cyber world is always changing, NIDS must adapt to that change and thus should be designed to learn from the environment it is placed in. Combined approaches that include different detection techniques are especially efficient, and this is due to the high level of versatility that can be helpful while dealing with new attacking types and techniques [25]. AI-powered Network Intrusion Detection Systems(NIDS), the workflow is shown in Figure 2(b), have revolutionized the field of

cybersecurity by providing a more proactive and adaptive approach to threat detection [26]. However, despite these advancements, several challenges remain that demand ongoing research.

Another major problem that can hinder the future development of AI-powered NIDS is the adversarial attacks that means that the data can be purposefully manipulated in front of the machine learning systems. These complex attacks result in reduction of false negatives and false positives which degrades the NIDS system [27]. It is, therefore, imperative to protect our AI cybersecurity systems from the above adversary tactics in order to preserve the reliability of the systems. Although deep learning models have fantastic capabilities in threat detection, they tend to be highly complicated and often referred to as 'black boxes', whereby the rationale used in making some decisions is hard to decipher. Improving the interpretability of these models is highly desirable to make it easier for consumers of these models to understand them and also to make sure that the decisions made by the AI systems do not have any biases [28]. The threats are dynamic in nature and therefore learning becomes part of NIDS as new data is collected. In traditional methods of machine learning, data needs to be retrained on updated databases, which is computationally costly and time-consuming. The future research should be oriented on the creation of the NIDS, which can learn in the real-time data streams or refresh the learnt information quickly enough to follow new threats. The overview of the layers utilized in the AI based NIDs system along with its description and processes is presented in Table 1.

Table 1. Layers and their description in AI based Intrusion detection system.

Layer	Description	Equations/Processes
Data Ingestion Layer	Collects and preprocesses network data.	$D_{\text{pre}} = f_{\text{preprocess}}(D_{\text{raw}})$
Machine Learning Layer	Employs supervised and unsupervised learning algorithms.	$M_{\text{supervised}} = \underbrace{\arg min}_{\theta} L_{\text{sup}}\left(D_{7_{\text{abelled}}}, \theta\right)$ $M_{\text{unsupervised}} = \underbrace{\arg min}_{\theta} L_{\text{unsup}}(D_{\text{raw}}, \theta)$
Deep Learning Integration	Integrates CNNs and RNNs for advanced pattern recognition.	$M_{\text{CNN}} = \underbrace{\arg min}_{\theta} L_{\text{CNN}}(D_{\text{raw}}, \theta)$ $M_{\text{RNN}} = \underbrace{\arg min}_{\theta} L_{\text{RNN}}(D_{\text{raw}}, \theta)$
Decision-Making Module	Processes machine learning outputs, triggers responses.	Decision $= f_{\text{decision}}\left(M_{\text{supervised}}, M_{\text{unsupervised}}, M_{\text{CNN}},\right.$
Continuous Learning Mechanism	Adapts algorithms in real-time to emerging threats.	$M'_{\text{supervised}} = M_{\text{supervised}} + \eta \nabla L_{\text{sup}}\left(D_{\text{new_labeled}}, \theta\right)$ $M'_{\text{unsupervised}} = M_{\text{unsupervised}} + \eta \nabla L_{\text{unsup}}\left(D_{\text{new_raw}}, \theta\right)$
Threat Intelligence Integration	Regularly updates the system with the latest threat intelligence feeds.	$\text{System}_{\text{updated}} = f_{\text{update_TI}}(\text{System}_{\text{current}}, \text{TI}_{\text{latest}})$
Adversarial Defense Measures	Mitigates adversarial threats using various techniques.	$M_{\text{adversarial}} = M_{\text{supervised}} + \eta_{\text{adv}} \nabla L_{\text{adv}}(D_{\text{adversarial}}, \theta)$ $\text{Detection}_{\text{adv}} = f_{\text{detect_adv}}\left(M_{\text{supervised}}, D_{\text{raw}}\right)$
User Interface	Provides an intuitive interface for cybersecurity professionals.	Interface $= f_{\text{interface}}(\text{Decision,Alerts,Reports})$

Notation: $\boldsymbol{D}_{\text{raw}}$: Raw input data, $\boldsymbol{D}_{\text{pre}}$: Preprocessed data, $\boldsymbol{D}_{\text{labeled}}$: Labeled data for supervised learning, $\boldsymbol{D}_{\text{new_labeled}}$: New labeled data for continuous learning, $\boldsymbol{D}_{\text{new_raw}}$: New raw data for continuous learning, $\boldsymbol{M}_{\text{supervised}}$: Supervised learning model, $\boldsymbol{M}_{\text{unsupervised}}$: Unsupervised learning model, $\boldsymbol{M}_{\text{CNN}}$: CNN model, $\boldsymbol{M}_{\text{RNN}}$: RNN model, $\boldsymbol{M}_{\text{adversarial}}$: Adversarially trained model, $\boldsymbol{L}_{\text{sup}}(\boldsymbol{D}, \boldsymbol{\theta})$: Loss function for supervised learning, where $\boldsymbol{\theta}$ represents the model parameters, $\boldsymbol{L}_{\text{unsup}}(\boldsymbol{D}, \boldsymbol{\theta})$: Loss function for unsupervised learning, $\boldsymbol{L}_{\text{CNN}}(\boldsymbol{D}, \boldsymbol{\theta})$: Loss function for CNN training, $\boldsymbol{L}_{\text{RNN}}(\boldsymbol{D}, \boldsymbol{\theta})$: Loss function for RNN training, $\boldsymbol{\nabla L}$: Gradient of the loss function, $\boldsymbol{\eta}$: Learning rate, $\boldsymbol{\eta}_{\text{adv}}$: Learning rate for adversarial training, $\boldsymbol{f}_{\text{decision}}$: Decisionmaking function based on inputs, $\boldsymbol{f}_{\text{update_TI}}$: Function to update the system with threat intelligence, $\boldsymbol{f}_{\text{detect_adv}}$: Function to detect adversarial patterns, $\boldsymbol{f}_{\text{interface}}$: Function to generate the user interface.

4. The proposed scheme of fortifying cyber defenses

The foundation of our proposed scheme lies in a robust and adaptive architecture designed to seamlessly integrate artificial intelligence into the fabric of network security [28]. At its core, the system comprises multiple layers, each playing a distinct role in fortifying cyber defenses.

- ***Data Sources and Capturing Techniques:*** Raw data are supplied from network traffic logs, sensors, and endpoint data as diverse sources for the NIDS. It monitors and analyzes the traffic on the network and seeks to discover cyclic activities that may point to the existence of threats.
- ***Feature Extraction and Preprocessing:*** Every attribute is carefully culled from the raw data and is preprocessed so that they can be fed to superior machine learning algorithms. This step is important so as to filter out extraneous information and format the rest in a way that will be most beneficial to the model[30]. The foundation of the NIDS is based on the complicated machine learning algorithms that can search and recognize threats. These algorithms, such as supervised, unsupervisor and deep learning models are used to identify both known and other potential threats with great accuracy.
- ***Machine Learning Models:*** Various machine learning models, such as supervised learning, unsupervised learning, and deep learning, are employed for threat detection.
- ***Threat Detection and Analysis:*** The system isolates the security threats and concerns, regard them as sensitive and scrutinizes them more thoroughly. This phase is extremely important in relation to the definition of threats in order to develop further specific and exhaustive investigations as well as the subsequent response.
- ***Alert Generation and Management:*** Once the system discovers possible threats, it produces notifications in order for the security personnel to take the necessary action and countermeasures. This makes threats to be handled as they come thus reducing the level of harm caused.
- ***Data Ingestion Layer:*** The Data Ingestion and Processing Layer of the system is the component that is responsible for the gathering of data from various sources as well as the subsequent implementation of enhancement and transformation procedures on the acquired data. This architecture is built in such a way that it has a strong data ingestion layer that is useful in processing an immense amount of network data [31].
- ***Machine Learning Layer:*** The machine learning layer is in fact the heart of the system, as it is indeed composed of a numerous group of various algorithms. Supervised model look for threats in the data which have been tagged so these algorithms look for known threats in data, while on the other hand unsupervised model works in real-time and first define the pattern and then look for its occurrence in data and if it finds any new pattern it considers it a threat and alerts the system.
- ***Deep Learning Integration:*** It is done by incorporating advanced deep learning models like Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN) to the system in order to have enhanced capability to recognize even subtle and sophisticated threats. Besides improving the detection possibility, the integration averts an increased likelihood of false-positive outcomes.
- ***Decision-Making Module:*** The decision-making module takes the outputs of the machine learning core and evaluates the level and types of threats. It then independently implements actions in accordance with the set policies thus guaranteeing that the appropriate responses are executed without much delay.

4.1. Continuous learning mechanisms

This being a reality, our NIDS is designed with the provision of continuous learning to adapt to ever improving cyber threats. This feature enables the system to learn with time new threats always trying to counter the existing system by challenging it with new ways, however, the system updates its algorithms and models in real time whenever new threats emerge [32].

4.2. Threat intelligence integration

Additionally, threat intelligence integration is integrated to the system to minimize the level of threats posed towards the system. Fresh feeds from the threat intelligence sources feed into the system to enhance its capability to detect the modern day threats with high levels of efficiency [33].

4.3. Adversarial defense measures

Since AI models used in such applications are highly sensitive to adversarial attacks, our NIDS countermeasures these threats. These are the adversarial training, the detection of adversarial patterns, and ways to improve model interpretability that strengthens the system against advanced abusive actions [34].

4.4. User-friendly interface and reporting

The NIDS also has an easy-to-use interface in order to help cybersecurity professionals manage and get the most out of the system. It has real-time widgets and alerts, an incident report, and more, so that the security team gets all the data in order to respond quickly to a threat.

5. Conclusion

This paper establishes that the adoption of AI, especially Machine Learning and Deep learning, has significantly altered the environment of NIDS. These autonomous systems which are equipped with artificial intelligence have the best ability in threat detection and protection as more advanced cyber threats arise all the time. It was possible to achieve sustainable goals in this area, where detection accuracy and efficiency have been significantly improved, thereby setting a high level of cybersecurity. Machine learning and deep learning combined have made it possible to have more active and interactive measures used to prevent and neutralize threats by evolving with the systems in relation to the cyber threats. Besides, the employment of multiple classifiers and associated blended techniques has enhanced the strength of NIDS, making them less susceptible to certain threats. However, issues like adversarial attacks, and handling deep-learning model interpretability as well as lifelong learning are still some of the most pressing concerns even today. However, there are certain challenges that need constant solving and improvement in the performance of AI-NIDS and the constant research and innovation towards the discovery of better ways in achieving the goals of this study cannot be overemphasized. NIDS has become a critical solution as cyber threats are still coming up with advanced methods of penetrating organizations' defenses. Thus, the future of the cybersecurity faces its proper evolution of highly intelligent and adaptive AI-based NIDS as the worldwide protection of the paramount digital infrastructures' safety and unblemished state.

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